MUKS ROBOTICS AI PVT. LTD.

TECHNICAL SPECIFICATION

SPACEO M1 – AI SOCIAL HUMANOID ROBOT

Document purpose: Technical specification for consideration in government / PSU / GeM tender submissions. The buyer's tender conditions and the final offered configuration shall take precedence.

1. Product Overview

Spaceo M1 is an AI-enabled social humanoid service robot designed for human interaction, visitor assistance, information delivery, guided navigation and public-facing service environments. It combines mobile navigation, camera-based perception, voice input/output, a touch display and articulated gestures. Use cases may include reception and visitor guidance, exhibitions, institutional information desks, hospitality and customer assistance.

2. Technical Specifications

Product / Model	Spaceo M1 – AI Social Humanoid Robot
Manufacturer / OEM	Muks Robotics AI Pvt. Ltd.
Robot type	Mobile social humanoid / service robot
Overall dimensions (H × W × L)	1625 × 420 × 390 mm
Weight (including battery)	Approximately 45 kg
Total degrees of freedom (DOF)	12 DOF
Arm degrees of freedom	5 DOF per arm
Base degrees of freedom	2 DOF
Maximum arm-joint torque	Approximately 160 kg·cm
Maximum travel speed	Up to 1.5 m/s
Arm length (upper arm + forearm)	Approximately 600 mm
Body construction	Aluminium body with polymer covers
Joint motors	Low-inertia, high-speed BLDC servo motors
Joint encoder	Single encoder per joint
Battery	25.6 V, 45 Ah $LiFePO_4$ battery
Microphone	4-microphone array
Speaker	5 W
Display	7-inch LED touch display
RGB camera	5 MP, Full HD
Depth camera	Intel RealSense depth camera
Navigation sensor	2D LiDAR
Wireless connectivity	Wi-Fi 6 and Bluetooth 5.2
Standard computing module	AMD Ryzen 7, 16 GB RAM, 500 GB SSD
AI computing module (configuration-dependent)	NVIDIA Jetson AGX Orin (approximately 275 TOPS) or RTX 5070 option, as offered
Cooling	Local air cooling
Electrical routing	Insulated harnessed routing
Indicative operating time	Approximately 8 hours; depends on operating conditions and configuration
Standby time	Up to approximately 72 hours, subject to configuration and conditions
Docking station	Included, as per offered package
Charger	Included
Manual remote control	Included
Over-the-air (OTA) software updates	Supported

Warranty	12 months

3. Functional Capabilities

Human–robot interaction through microphone array, speaker and touch display.

Voice-based interaction and conversational assistance, subject to the installed AI software, language pack and network / compute configuration.

Autonomous mobile navigation and obstacle-aware movement using onboard navigation sensors, subject to site mapping and commissioning.

Visual perception using RGB and depth-camera sensors.

Expressive arm gestures and articulated movements for interactive demonstrations and service workflows.

Configurable information delivery and domain-specific content for reception, visitor guidance, product / facility information and exhibitions.

Remote/manual control for supervised operation, where supplied in the offered configuration.

Software update capability through OTA updates, subject to network access and the OEM's update process.

4. Standard Supply Scope

One (1) Spaceo M1 robot in the configuration quoted in the commercial bid.

Battery and compatible battery charger.

Docking station, where specified in the offered configuration.

Manual remote control.

User / operating documentation.

Basic installation, commissioning and operator handover, if included in the tender scope or purchase order.

Warranty support for 12 months from the date defined in the purchase order / contract.

5. Suggested Acceptance and Demonstration Tests

Physical inspection of supplied robot, accessories, model and serial number.

Power-on, charging and battery status demonstration.

Demonstration of forward, reverse and turning movement in a safe, designated area.

Demonstration of navigation and obstacle response in a representative site environment, subject to safety conditions.

Demonstration of camera, microphone, speaker and touch display.

Demonstration of the agreed conversational / information-assistance workflow and configured language(s).

Demonstration of arm gestures and any agreed interaction sequence.

Handover of user documentation and explanation of warranty / support process.

6. Tender Compliance Notes

The dimensions, weight, DOF, sensors and other numeric values above are based on the Spaceo M1 specification published by Muks Robotics at the time of preparation.

The AI computing option is configuration-dependent. The exact compute module supplied must be identified in the technical bid / quotation before submission.

Operating time and standby time are indicative and may vary with navigation activity, workload, battery condition, environment and installed configuration.

Do not treat payload capacity, lifting capacity, language count, facial-recognition functionality, cybersecurity certifications, IP rating, safety certifications or other unlisted values as guaranteed unless expressly confirmed in the final OEM offer.

The buyer should evaluate the robot against the actual use case and tender requirements. Any mandatory specification not covered here should be confirmed in writing by the OEM before bid submission.

If a tender requires standards / certifications, attach only valid certificates applicable to the exact offered model and configuration.

7. OEM Details

Muks Robotics AI Pvt. Ltd.

Office No. 405, Maruti Millennium Tower, Maruti Chowk, Pune–Bangalore Highway, Pashan Exit, next to Tata Showroom, Baner, Pune, Maharashtra – 411045.

Website: www.muksrobotics.com

Email: shivam.kelkar@muksrobotics.com

Authorized Signatory: Dr. Mukesh Bangar
Designation: Founder / Director
For: Muks Robotics AI Pvt. Ltd.
Signature & Company Seal: _____
Date: _____ Place: Pune

Reference

Official product page: https://muksrobotics.com/spaceo-m1

Important: This document is a draft technical specification for tender preparation. Please verify the offered configuration, commercial scope and any tender-specific compliance before signing or uploading.